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1-1 PRIVATE TUTORS

ALGEBRA

Rearranging Formulae

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KEY WORDS formula *(tātai)* · subject · rearrange *(whakaraupapa anō)* · inverse operation *(mahi kōaro)*

A Inverse operations ●○○

Remember: An inverse operation undoes another operation. Subtracting 5 undoes adding 5, and dividing by 3 undoes multiplying by 3.

1. Write the operation that undoes each one.

a) $+5$ _____ **b)** $\times 3$ _____ **c)** -8 _____ **d)** $\div 4$ _____

e) $+a$ _____ **f)** $\times k$ _____ **g)** $-w$ _____ **h)** $\div t$ _____

B Rearranging in one step ●●○

Remember: The subject is the letter on its own. To change the subject, undo what has been done to the new letter, on both sides: $y = x + 5$ becomes $x = y - 5$.

2. Make the letter in square brackets the subject of each formula.

a) $y = x + 5 \quad [x]$ **b)** $y = x - 7 \quad [x]$ **c)** $y = 4x \quad [x]$

d) $y = \frac{x}{3} \quad [x]$ **e)** $A = lw \quad [l]$ **f)** $d = st \quad [t]$

g) $P = 4s \quad [s]$ **h)** $v = u + t \quad [u]$ **i)** $T = c - 4 \quad [c]$

j) $C = \pi d \quad [d]$ **k)** $s = \frac{d}{t} \quad [d]$ **l)** $p = q + r \quad [q]$

C Rearranging in two steps ●●●

Watch out: Undo the operations in the reverse order. For $y = 2x + 3$, subtract 3 first, then divide by 2: $x = \frac{y - 3}{2}$.

3. Make the letter in square brackets the subject of each formula.

a) $y = 2x + 3 \quad [x]$ **b)** $y = 3x - 2 \quad [x]$ **c)** $C = 5n + 20 \quad [n]$

d) $v = u + at \quad [t]$ **e)** $P = 2l + 2w \quad [l]$ **f)** $y = \frac{x}{4} + 1 \quad [x]$

g) $A = \frac{bh}{2} \quad [h]$ **h)** $y = 3(x + 1) \quad [x]$ **i)** $T = 15 - 2h \quad [h]$

j) $y = 5x + 4 \quad [x]$ **k)** $k = 3m - 6 \quad [m]$ **l)** $y = \frac{x}{2} - 3 \quad [x]$

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D Using a rearranged formula ●●●

4. The distance travelled at a steady speed is given by $d = st$.

a) Make t the subject of the formula.

b) Find t when $d = 120$ and $s = 40$.

5. The area of a rectangle is given by $A = lw$.

a) Make w the subject of the formula.

b) Find w when $A = 48$ and $l = 8$.

6. The cost in dollars of hiring a hall for n hours is given by $C = 5n + 20$.

a) Make n the subject of the formula.

b) Find n when $C = 65$.

7. The speed of a car that is speeding up is given by $v = u + at$.

a) Make t the subject of the formula.

b) Find t when $v = 17$, $u = 5$ and $a = 3$.

8. The perimeter of a rectangle is given by $P = 2l + 2w$.

a) Make l the subject of the formula.

b) Find l when $P = 30$ and $w = 6$.

★ Extension

9. Make the letter in square brackets the subject of each formula.

a) $A = \frac{1}{2}(a + b)h \quad [h]$

b) $y = \frac{x + 3}{2} \quad [x]$

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Rearranging Formulae

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS formula *(tātai)* · subject · rearrange *(whakaraupapa anō)* · inverse operation *(mahi kōaro)*

A Inverse operations ●○○

Remember: An inverse operation undoes another operation. Subtracting 5 undoes adding 5, and dividing by 3 undoes multiplying by 3.

1. Write the operation that undoes each one.

a) $+7$ _____ **b)** $\times 2$ _____ **c)** -6 _____ **d)** $\div 5$ _____

e) $+b$ _____ **f)** $\times m$ _____ **g)** $-h$ _____ **h)** $\div s$ _____

B Rearranging in one step ●●○

Remember: The subject is the letter on its own. To change the subject, undo what has been done to the new letter, on both sides: $y = x + 5$ becomes $x = y - 5$.

2. Make the letter in square brackets the subject of each formula.

a) $y = x + 8 \quad [x]$ **b)** $y = x - 3 \quad [x]$ **c)** $y = 6x \quad [x]$

d) $y = \frac{x}{5} \quad [x]$ **e)** $A = bh \quad [b]$ **f)** $d = st \quad [s]$

g) $P = 3s \quad [s]$ **h)** $v = u + t \quad [t]$ **i)** $T = c - 9 \quad [c]$

j) $C = \pi d \quad [d]$ **k)** $m = \frac{k}{n} \quad [k]$ **l)** $a = b + c \quad [b]$

C Rearranging in two steps ●●●

Watch out: Undo the operations in the reverse order. For $y = 2x + 3$, subtract 3 first, then divide by 2: $x = \frac{y - 3}{2}$.

3. Make the letter in square brackets the subject of each formula.

a) $y = 4x + 1 \quad [x]$ **b)** $y = 2x - 5 \quad [x]$ **c)** $C = 8n + 15 \quad [n]$

d) $v = u + at \quad [a]$ **e)** $P = 2l + 2w \quad [w]$ **f)** $y = \frac{x}{3} + 2 \quad [x]$

g) $A = \frac{bh}{2} \quad [b]$ **h)** $y = 2(x + 5) \quad [x]$ **i)** $T = 12 - 2h \quad [h]$

j) $y = 6x + 5 \quad [x]$ **k)** $k = 4m - 8 \quad [m]$ **l)** $y = \frac{x}{5} - 2 \quad [x]$

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D Using a rearranged formula ●●●

4. The distance travelled at a steady speed is given by $d = st$.

a) Make s the subject of the formula.

b) Find s when $d = 150$ and $t = 3$.

5. The area of a rectangle is given by $A = lw$.

a) Make l the subject of the formula.

b) Find l when $A = 54$ and $w = 6$.

6. The cost in dollars of hiring a boat for n hours is given by $C = 8n + 15$.

a) Make n the subject of the formula.

b) Find n when $C = 71$.

7. The speed of a car that is speeding up is given by $v = u + at$.

a) Make a the subject of the formula.

b) Find a when $v = 26$, $u = 6$ and $t = 5$.

8. The perimeter of a rectangle is given by $P = 2l + 2w$.

a) Make w the subject of the formula.

b) Find w when $P = 34$ and $l = 9$.

★ Extension

9. Make the letter in square brackets the subject of each formula.

a) $m = \dfrac{a + b}{2}$ $\quad [a]$

b) $y = \dfrac{x - 4}{3}$ $\quad [x]$

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